NexusSOP

Project Analysis, Completion Status and Judge Presentation Guide

Problem Statement 2: Multimodal GenAI Assistant for Industrial SOPs, Manuals and Expert Knowledge

Prepared from the recent project review and implementation discussions

Current validation: **87 automated tests passed**

1. Executive Summary

NexusSOP is a safety-aware industrial knowledge assistant. It searches approved manuals and SOPs, returns evidence-backed answers, generates structured procedures, displays original figures, supports voice interaction, preserves engineer feedback, and refuses to guess when evidence is insufficient.

Estimated hackathon prototype completion: 75–85%. The core product is implemented. The remaining work is concentrated in deeper visual reasoning, evaluation metrics, cloud deployment demonstration, and final presentation polish.

> **Core positioning:** NexusSOP is not a generic chatbot over PDFs. It is a governed industrial knowledge workflow with approval, citations, abstention, escalation and human review.

Recommended one-line pitch:

> **“NexusSOP transforms approved industrial manuals and expert knowledge into traceable, voice-accessible SOPs while refusing to hallucinate unsafe instructions.”**

2. Alignment with Problem Statement 2

Requirement	Status	Evidence in project
GenAI assistant	Implemented	Optional NVIDIA/Gemini model with local fallback.
RAG over manuals/SOPs	Implemented	Approved-document retrieval, relevance scoring and citations.
Text documents	Implemented	PDF, DOCX, TXT and Markdown ingestion.
Images and diagrams	Partially implemented	OCR, embedded image extraction, page snapshots and original figure export.
Grounded answers	Strong	Retrieved references are passed as numbered evidence and citations are validated.
Troubleshooting/SOP guidance	Implemented	Structured steps, cautions, checks, figures and PDF export.
Chat interface	Implemented	Browser chat with persistent sessions.
Voice interface	Implemented foundation	Gemini Live relay and server-side knowledge tool.
Expert knowledge	Implemented	Engineer feedback, notes, shared rooms and synthesized notes.
Analytics	Implemented	Dashboard, query outcomes, escalations and CSV export.
Cloud scalability	Partially implemented	Docker and modular FastAPI architecture; current default is local SQLite.

3. Completed Features

- Multi-user authentication with admin and engineer roles.
- Session cookies, password hashing, invitations and user management.

• Document upload with human approval before retrieval.

• PDF, DOCX, TXT, Markdown and image ingestion.

• OCR preprocessing for image uploads.

• PDF page markers and embedded image/figure extraction.

• Lexical RAG with stemming, IDF relevance and query-term coverage.

• Retrieved evidence passed to the model as numbered references.

• Citation validation so unsupported source IDs are removed.

• Low-evidence abstention and safety-oriented operational refusal.

• Chat, shared rooms, persistent text sessions and collaborative polling.

• Voice relay with server-side knowledge retrieval.

• SOP generation with steps, cautions, checks, citations and figures.

• Engineer feedback, manual notes and synthesized notes.

• Executive PDF report generation with source reference pages.

• New compact black-and-white Quick B/W SOP export.

• Analytics dashboard, escalation tickets and CSV export.

• Docker setup and automated test coverage.

4. Recent Enhancement: Quick B/W SOP

A separate compact export mode was added without removing the existing executive report. This directly supports the request for a simple black-and-white sheet containing less text, important content, diagrams and original-page references.

Feature	Implementation
Black-and-white layout	A new compact ReportLab PDF renderer uses a clean monochrome layout.
Less text	Step descriptions and warnings are shortened to essential content.
Important information	Essential steps, cautions, warnings and verification checks are separated.
Original diagrams	Only source-linked extracted figures are included; no invented diagrams are used.
Original page references	Citation IDs are resolved to page markers such as “p. 12”.
UI access	A “Quick B/W SOP” button appears beside the executive PDF export.
Endpoint	Authenticated GET/POST endpoint: /api/sops/{id}/quick-export

Judge value: The executive report is useful for supervisors and audit review, while the Quick B/W SOP is useful for a technician who needs a fast, printable reference.

5. Pending and Partially Complete Work

Priority	Pending item	Why it matters
High	True diagram understanding	The assistant should reason about labels, symbols, arrows, tables and schematics, not only extract/display images.
High	Polished voice-to-SOP scenario	Show voice symptom/error input, clarification, retrieval, SOP generation and display as one complete flow.
High	Evaluation benchmark	Measure retrieval hit rate, citation precision, abstention accuracy, latency and escalation rate.
High	Cloud deployment story	Demonstrate Docker/cloud architecture, object storage and a database separation plan.
Medium	Multilingual support	Hindi or another regional-language mode could make the solution more memorable and inclusive.
Medium	Production security	Encryption at rest, malware scanning, secure secrets, audit retention and removal of default credentials.
Medium	Industrial integrations	CMMS/ERP/maintenance ticket connectors and external alert delivery.
Medium	Advanced analytics	Time saved, common faults, unanswered questions, manual usage and escalation trends.
Roadmap	Expert safety validation	Domain-reviewed test sets and formal approval before real equipment use.

Important: These items are not all required for the hackathon prototype. For judging, prioritize the first four high-impact items and present the rest as a production roadmap.

6. Features Most Likely to Attract Judges

• **Evidence-first answers:** Every answer points back to exact approved source material.

• **Safety-aware abstention:** The assistant refuses unsupported or dangerous machine-operation requests.

• **Human-in-the-loop governance:** New documents require approval before entering retrieval.

• **Multimodal foundation:** Manuals, images, OCR, figures and voice are included in one workflow.

• **Automatic SOP generation:** Evidence becomes structured steps, cautions, checks and printable outputs.

• **Expert knowledge preservation:** Feedback and field notes remain attached to the generated SOP.

• **Two audience outputs:** Detailed executive report for supervisors and compact B/W sheet for technicians.

• **Offline fallback:** Local extractive RAG keeps the demo usable even without an external model key.

• **Analytics:** Supervisors can see recurring issues, outcomes and escalations.

• **Scalable foundation:** FastAPI, Docker and modular ingestion provide a clear path to cloud deployment.

7. Recommended Live Demonstration

Use one complete industrial story instead of showing disconnected features:

Step	What to show	Judge takeaway
1	Upload a machine manual and show Pending Review.	Knowledge is governed, not blindly indexed.
2	Approve the manual as administrator.	Human review controls source quality.
3	Ask a grounded question and open the cited page.	The answer is traceable.
4	Ask a question absent from the manual.	The assistant abstains rather than inventing.
5	Ask a risky operational question.	Safety boundary and escalation are visible.
6	Use voice to provide machine, symptom and error code.	Factory-floor interaction is practical.
7	Generate an SOP with figures and checks.	RAG becomes a usable work artifact.
8	Download Quick B/W SOP.	Technicians get a concise printable reference.
9	Add engineer feedback and show analytics.	Expert knowledge compounds over time.

8. Talking Points for Judges

Problem: Industrial teams lose time searching long manuals, while unsupported AI instructions can create safety risk.

Solution: NexusSOP retrieves only approved evidence, shows the source, generates concise procedures and knows when to stop.

Innovation: The project combines RAG, governance, voice, figures, feedback and analytics rather than presenting only a chat window.

Impact: It can reduce manual search time, preserve retiring experts' knowledge and route uncertain cases to qualified personnel.

Trust: The system's most important behavior is not just answering; it is refusing unsupported answers and exposing its evidence.

Scalability: Additional machines, manuals, plants, languages and maintenance systems can be onboarded through the modular architecture.

Suggested judge-facing answer when asked about limitations:

> "This is a hackathon prototype, not a certified plant-control system. We deliberately show that limitation. The next validation stage is domain-expert evaluation with authorized manuals, measurable retrieval and refusal tests, secure cloud storage and approved maintenance-system integration."

9. Technical Architecture Summary

• Browser interface: industrial chat, dashboard, document review, SOP cards and voice pages.

• FastAPI backend: authentication, document APIs, chat, voice relay, SOP APIs and analytics.

• Ingestion layer: PyMuPDF, pypdf, python-docx, image normalization and Tesseract OCR.

• Retrieval layer: Porter stemming, lexical scoring, IDF relevance, evidence coverage and source excerpts.

• Generation layer: optional NVIDIA/Gemini models with extractive fallback.

• Governance layer: role-based access, approval gates, citation filtering, abstention and escalation.

• Persistence: SQLite records for users, sessions, rooms, documents, assets, SOPs, feedback, notes and analytics.

• Export layer: executive dark report and compact black-and-white source-referenced SOP.

10. Validation and Current Confidence

Check	Result
Automated test suite	87 passed, 1 non-blocking fitz deprecation warning.
Python compilation	Passed after compact export changes.
Patch formatting	git diff --check passed.
Compact export integration test	Passed; PDF contains essential steps and source-page references.
Existing executive export	Preserved and regression-tested.

The project is suitable for a strong hackathon demonstration. Avoid claiming measured factory savings until authorized plant data and expert-reviewed evaluation are available.

11. Final Presentation Checklist

• Prepare one approved manual and one realistic machine/error scenario.

• Verify all demo API keys, login credentials and uploaded documents before judging.

• Do not use the default admin password in the final presentation.

• Keep the cited original manual page ready to open after every key answer.

• Show both executive PDF and Quick B/W SOP exports.

• Demonstrate one successful grounded answer and one safe abstention.

• Explain current limitations honestly and connect them to the roadmap.

• End with the safety-aware positioning statement, not with a list of technologies.

> **Closing statement:** NexusSOP makes industrial knowledge usable, traceable and safer by connecting approved manuals, expert review, multimodal interaction and concise field-ready SOPs.